THRIVEHEALTH360 · THE METABOLIC RESET

The Full Guide

Six supplement topics — what the research shows, where it is limited, and what to discuss with your doctor — with 30 cited sources.

6	8	30	12
Supplement Topics	Total Chapters	Cited Sources	Week Reset

Ayman & The ThriveHealth360 Editorial Team · Revised October 2026

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Table of Contents

Intro **The Metabolic Awakening**
Why insulin resistance is the real enemy

Ch. 1 **The Berberine Protocol**
What the research says about berberine · 5 refs

Ch. 2 **The Receptor Gatekeepers**
Magnesium Bisglycinate & Chromium Picolinate · 5 refs

Ch. 3 **Alpha-Lipoic Acid**
What the evidence shows · 5 refs

Ch. 4 **Omega-3 & Meta-inflammation**
What omega-3 can and cannot do · 4 refs

Ch. 5 **Vitamin D3 & K2**
Vitamin D in prediabetes, and where K2 fits · 5 refs

Ch. 6 **Akkermansia muciniphila**
Early evidence for a gut-microbe supplement · 5 refs

Concl. **The 12-Week Reset Calendar**
Full protocol sequencing blueprint

Append. **Master Bibliography**
30 cited sources with DOI / PMID / PMC where available

Append. **Medical Disclaimer & Disclosure**

Key Terms Glossary

AMPK	AMP-activated protein kinase — master cellular energy sensor. When activated it boosts fat-burning and glucose uptake via GLUT4 upregulation.
GLUT4	Glucose Transporter Type 4 — the protein that physically moves glucose from blood into muscle cells when AMPK is active.
Tyrosine Kinase	The enzyme that activates the insulin receptor after insulin binds. Requires Magnesium as an essential cofactor to function.
Chromodulin	A chromium-binding oligopeptide proposed to amplify the insulin signal at the receptor. Its role in humans is still debated.
SPMs	Specialized Pro-resolving Mediators (resolvins, protectins, maresins) — lipid molecules made from omega-3 EPA/DHA that are thought to help switch off inflammation. Whether supplements raise them meaningfully in people is still being studied.

Secosteroid	What Vitamin D3 truly is — a secosteroid hormone that acts on a receptor found in many tissues, including the pancreas.
Endotoxemia	The state where bacterial LPS leaks through a thin gut lining into the bloodstream, driving systemic meta-inflammation and insulin resistance.
Amuc_1100	A surface protein on Akkermansia muciniphila that stimulates GLP-1 secretion from intestinal L-cells. Heat-treatment (pasteurization) stabilizes it.

INTRODUCTION

The Metabolic Awakening

Why insulin resistance is the real enemy

> *"Food, movement, sleep and stress come first. This guide looks at what the research says about six supplements, including where the evidence is limited."*

If you are reading this, you may be one of the many adults whose cells respond less well to insulin — a state called **Insulin Resistance**. For decades, the standard advice has been 'eat less and move more.' Food choices and movement matter a great deal, and this guide does not replace them. It looks at one further question: whether certain supplements add anything for people with insulin resistance.

This guide looks at six supplement topics and what the research shows for each, including where it is limited. Each chapter ends with its sources, and the final chapter shows one way to sequence them over 12 weeks. It is educational information, not medical advice, and it is not a substitute for food, movement, sleep, stress care or your doctor's treatment.

The Six Topics

Ch. 1	**Berberine**	A plant compound studied for blood sugar
Ch. 2	**Magnesium & Chromium**	Minerals involved in insulin signaling
Ch. 3	**Alpha-Lipoic Acid**	Studied mainly for diabetic neuropathy
Ch. 4	**Omega-3 EPA/DHA**	Fish-oil fats and inflammation markers
Ch. 5	**Vitamin D3 & K2**	Vitamin D in prediabetes
Ch. 6	**Akkermansia**	An early-stage gut-microbe supplement

Each topic involves a different biological pathway. The calendar in the final chapter shows one way to introduce them gradually; it is an example, not a prescription. The strength of the evidence differs a lot between topics, and none of these supplements replaces food, movement, sleep or medical care.

CHAPTER 1 · NUTRIENT RESEARCH

CHAPTER 1

Berberine: What the Research Shows

A plant compound studied for blood sugar — what the trials show, and what they do not.

Berberine is a plant alkaloid found in Berberis species, and it has attracted more research interest than most supplements studied for blood sugar. A 2024 meta-analysis of randomized trials in people with type 2 diabetes reported lower fasting and post-meal glucose with berberine than with placebo or control. These were people who already have diabetes, and berberine is not a substitute for prescribed medication such as metformin. [1]

01. The AMPK Master Switch

A leading proposed mechanism for berberine is activation of the **AMPK pathway**. Think of AMPK as your body's master energy sensor. When 'off,' your body stores fat and traps sugar in the bloodstream. In laboratory and animal studies, berberine switching it 'on' increases GLUT4 transporters, which helps move glucose into muscle tissue. [5]

WHAT THE YIN 2008 PILOT FOUND (SUMMARY)

"In a small pilot study of people with type 2 diabetes, berberine lowered blood glucose and HbA1c over three months, and performed similarly to metformin in the head-to-head comparison."

— Summarised from Yin J, et al. Metabolism 2008 (pilot study) | PMID 18442638

02. Can Milk Thistle Help Absorption?

Berberine is poorly absorbed because of first-pass metabolism and P-glycoprotein efflux. **Silymarin** (milk thistle) has been proposed to help by inhibiting that efflux, and some products combine the two. A meta-analysis of five placebo-controlled trials (497 people) reported improvements in cholesterol, triglycerides and fasting glucose with the berberine–silymarin combination compared with placebo. It did not measure berberine levels in the blood and did not compare the combination with berberine alone, so the idea that silymarin doubles absorption is not established. [4] Some studies give berberine before meals; whether that timing matters is unclear.

THE SILYMARIN CARRIER

Berberine + Milk Thistle

Proposed to improve absorption. The meta-analysis (Fogacci 2019) compared clinical outcomes with placebo; it did not measure absorption. [4]

MEAL TIMING

Before or With Meals

A common schedule in studies; whether timing matters is unclear. Three equally spaced doses a day is the usual pattern.

03. What Studies Typically Used

Total Daily Amount	**up to 1,500 mg (studies vary)**
Dosing Frequency	**2-3 divided doses is common**
Meal Timing	**Before or with meals; timing not established**
Study Duration	**Most trials ran 8-12 weeks**
Quality to Look For	**Third-party tested (e.g. NSF, USP)**

⚠ CRITICAL MEDICAL WARNING

Berberine can lower blood glucose. If you are prescribed glucose-lowering medication or insulin, you MUST consult your doctor before starting it. Combining it with diabetes medication can cause **dangerous hypoglycemia**. Monitor your fasting glucose closely during the first 2 weeks, and stop and contact your doctor if you feel shaky, dizzy, or unusually hungry.

PEER-REVIEWED REFERENCES · PUBMED / PMC VERIFIED

1. Wang J, Bi C, Xi H, Wei F. (2024). Effects of Administering Berberine Alone or in Combination on Type 2 Diabetes: A Systematic Review and Meta-Analysis. *Frontiers in Pharmacology.* DOI: 10.3389/fphar.2024.1455534

PubMed · PMID 39640489 ■

50-study meta-analysis (n=4,150): Berberine alone significantly reduced FPG (MD=−0.59 mmol/L), 2h postprandial glucose (MD=−1.57 mmol/L), LDL-C, TC, and TG vs. placebo.

2. Liu D, et al. (2025). Efficacy and Safety of Berberine on the Components of Metabolic Syndrome. *Frontiers in Endocrinology.* DOI: 10.3389/fendo.2025.1568441

PMC Free Full Text · PMC12307485 ■

12-RCT meta-analysis confirming Berberine improved all core metabolic syndrome components including fasting glucose, triglycerides, and blood pressure.

3. Yin J, Xing H, Ye J. (2008). Efficacy of Berberine in Patients with Type 2 Diabetes Mellitus. *Metabolism.* 57(5):712–7. DOI: 10.1016/j.metabol.2008.01.013

PubMed · PMID 18442638 ■ | PMC Free Full Text · PMC2410097 ■

Foundational RCT: Berberine reduced HbA1c by 2%, FPG by 1.84 mmol/L — comparable to metformin and rosiglitazone with no serious adverse effects.

4. Fogacci F, Tocci G, Presta V, et al. (2019). Metabolic Effect of Berberine-Silymarin Association: A Meta-Analysis of Randomized Controlled Trials. *Nutrients.* 11(11):2704. DOI: 10.3390/nu11112704

PubMed · PMID 31717839 ■

Meta-analysis: Berberine+Silymarin significantly improved glycemic and lipid parameters beyond Berberine alone, confirming the bio-enhancer synergy mechanism.

5. Wang Y, et al. (2022). Berberine and Its Role in Activation of AMPK: A Review of Metabolic Implications. *Frontiers in Pharmacology.* DOI: 10.3389/fphar.2022.936393

PubMed · PMID 36091215 ■

Mechanistic review confirming AMPK activation as Berberine's primary metabolic pathway, with downstream effects on GLUT4 upregulation and hepatic gluconeogenesis inhibition.

CHAPTER 2 · NUTRIENT RESEARCH

CHAPTER 2

Magnesium & Chromium: What the Research Shows

Two minerals studied alongside insulin signaling — and what the evidence actually supports.

Insulin acts like a key that lets glucose into your cells. In insulin resistance, that process works less well, and cells take up less glucose for the same amount of insulin. Magnesium and chromium are minerals involved in parts of this process, and researchers have studied whether supplementing them helps.

01. Magnesium

Magnesium is a cofactor for the enzyme **tyrosine kinase**, which the insulin receptor needs to pass on its signal, so low magnesium is one plausible way insulin signaling could be affected. Low magnesium intake is also linked with higher type 2 diabetes risk in observational studies. A 2025 review of 23 randomized trials (1,345 people with type 2 diabetes) found magnesium supplements modestly lowered fasting glucose, though the authors describe the effect on long-term control as limited. [1] Most of this evidence is in people who already have diabetes, not prediabetes.

WHAT THE ASBAGHI-STYLE REVIEW FOUND (SUMMARY)

"Magnesium supplements modestly reduce fasting glucose in people with type 2 diabetes; the effect on long-term glycaemic control appears limited, and the evidence is of mixed quality."

— Summarised from a 2025 systematic review of 23 RCTs (Sultan Qaboos Univ Med J), PMC12244252

02. Chromium Picolinate

Chromium has been studied for decades as a possible aid to insulin signaling, partly through a proposed binding protein called chromodulin, though its role in humans is still debated. [5] A 2007 systematic review found chromium lowered HbA1c and fasting glucose in people with diabetes, but found no benefit in people without diabetes, and rated the overall evidence quality as limited.

PHASE 1 – CATALYST

Magnesium Bisglycinate

Bisglycinate is generally gentle on the stomach compared with magnesium oxide. Limit elemental magnesium from supplements to 350 mg/day unless your doctor advises otherwise.

up to 350 mg elemental / day

PHASE 2 – AMPLIFIER

Chromium Picolinate

A commonly studied form. Evidence of benefit is limited outside people who already have diabetes, and is weak for prediabetes specifically.

200–600 mcg / day

⚠ NUTRIENT COMPETITION WARNING

Chromium Picolinate and Iron compete for the same transport protein (transferrin). If you are taking an iron supplement or iron-fortified multivitamin, ensure at least a 4-hour window between doses.

PEER-REVIEWED REFERENCES · PUBMED / PMC VERIFIED

1. Multiple authors (2025). Effect of Magnesium Supplements on Improving Glucose Control, Blood Pressure and Lipid Profile in Patients With Type 2 Diabetes Mellitus: A Systematic Review and Meta-Analysis. *Sultan Qaboos University Medical Journal* 25(1):382-94.

PMC Free Full Text · PMC12244252 ■

23 RCTs, 1,345 people with type 2 diabetes: magnesium modestly reduced fasting glucose; the authors describe the effect on long-term glycaemic control as limited.

2. Veronese N, et al. (2021). Oral Magnesium Supplementation for Treating Glucose Metabolism Parameters in People with or at Risk of Diabetes. *Nutrients.* DOI: 10.3390/nu13114074

PubMed · PMC8619199 ■

25 RCTs, 1,677 people: in diabetes, a small reduction in fasting glucose with no effect on HbA1c, fasting insulin or HOMA-IR; at high risk of diabetes, small improvements in fasting glucose, HOMA-IR and 2-hour OGTT glucose, again with no HbA1c or insulin effect.

3. Simental-Mendía LE, Sahebkar A, Rodríguez-Morán M, Guerrero-Romero F. (2016). A Systematic Review and Meta-Analysis of Randomized Controlled Trials on the Effects of Magnesium Supplementation on Insulin Sensitivity and Glucose Control. *Pharmacological Research* 111:272-282.

DOI 10.1016/j.phrs.2016.06.019 ■

Magnesium taken for four months or longer improved fasting glucose and insulin resistance in people with and without diabetes; shorter trials did not show the same effect. A meta-analysis shows an association, not a mechanism.

4. Barbagallo M, Veronese N, Dominguez LJ. (2022). Magnesium in Type 2 Diabetes Mellitus, Obesity, and Metabolic Syndrome. *Nutrients* 14(3):714.

DOI 10.3390/nu14030714 ■

Narrative review (not a systematic review) discussing links between low magnesium and glucose, blood pressure and lipid measures.

5. Balk EM, et al. (2007). Effect of Chromium Supplementation on Glucose Metabolism and Lipids: A Systematic Review of Randomized Controlled Trials. *Diabetes Care* 30(8):2154-2163.

PubMed · PMID 17519436 ■

In people with diabetes, chromium lowered HbA1c by about 0.6 percentage points and fasting glucose by about 1.0 mmol/L; no benefit was seen in people without diabetes, and study quality was often poor.

CHAPTER 3

Alpha-Lipoic Acid: What the Research Shows

Mainly studied in diabetic neuropathy — the evidence is mixed, and it is not a glucose-lowering therapy.

Persistently high blood sugar can damage peripheral nerves over time, partly through processes involving **Advanced Glycation End Products (AGEs)** and oxidative stress. Alpha-lipoic acid (ALA) is an antioxidant that has been studied as a treatment for people who already have diabetic peripheral neuropathy, not as a way to lower blood sugar. The evidence on whether it helps symptoms is mixed: a 2024 Cochrane review found it probably has little or no effect on neuropathy symptoms, while a separate 2025 meta-analysis found it may help. [1]

01. Why ALA Is Studied

ALA is **amphipathic** — it dissolves in both water and fat, unlike vitamin C (water only) or vitamin E (fat only), which is part of why researchers have been interested in it as an antioxidant that could reach nerve tissue. This is a property observed in the lab; it does not by itself prove a clinical benefit. [2]

WHAT THE COCHRANE REVIEW FOUND (SUMMARY)

"In people with diabetic neuropathy, oral alpha-lipoic acid probably has little or no effect on neuropathy symptoms at six months. Doses studied were 600–1,800 mg a day."

— Summarised from Baicus C, et al. Cochrane Database Syst Rev 2024 | PMID 38205823

02. R-ALA vs. Standard ALA

Most ALA sold is a **racemic** mixture: about half R-ALA (the form the body makes naturally) and half S-ALA (a synthetic byproduct). R-ALA is sometimes sold on its own at a premium, but the trials behind ALA's evidence base mostly used the standard racemic form, so it's unclear whether R-ALA alone performs any better.

THERAPEUTIC DOSING

Active Nerve Discomfort

A dose range used in some trials for people with diagnosed neuropathy symptoms. Use only under your doctor's guidance; higher doses have more side effects.

up to 1,200 mg / day, split doses

PREVENTIVE DOSING

Microvascular Protection

A commonly studied dose. There is no established evidence that ALA prevents neuropathy before symptoms appear.

300–600 mg / day

⚠ WARNING: THIAMINE (B1) DEPLETION

ALA can compete for the same cellular uptake pathways as Vitamin B1 (Thiamine). For long-term use exceeding 12 weeks, ask your doctor whether a B1 (thiamine) supplement is appropriate for you during long-term useions.

PEER-REVIEWED REFERENCES · PUBMED / PMC VERIFIED

1. Baicus C, Purcarea A, von Elm E, Delcea C, Furtunescu FL. (2024). Alpha-Lipoic Acid for Diabetic Peripheral Neuropathy. *Cochrane Database of Systematic Reviews* 1:CD012967.

PubMed · PMID 38205823 ■

In people with diabetic neuropathy, oral ALA (doses 600-1,800 mg/day) probably has little or no effect on symptoms at 6 months, and may have little or no effect on nerve impairment. One of two trials included showed no effective treatment for DPN existed at the time.

2. Papanas N, Papachristou S, Ziegler D. (2025). Alpha-Lipoic Acid in Diabetic Peripheral Neuropathy: Addressing the Challenges and Complexities Surrounding a 70-Year-Old Compound. *Current Issues in Molecular Biology* 47(6):402.

PMC Free Full Text · PMC12191796 ■

Narrative review of ALA for diabetic peripheral neuropathy; notes current DPN therapies mainly offer symptom relief rather than a cure.

3. Das S, Viswanatha GL, Nandakumar K, Kishore A, Hanumanthappa S. (2025). Alpha Lipoic Acid: Advancing Insights in Diabetic Neuropathy Through Updated Systematic Review and Meta-Analysis. *Exploration of Neuroprotective Therapy* 5:1004125.

DOI 10.37349/ent.2025.1004125 ■

9 RCTs, 1,345 people with diabetic neuropathy: oral ALA (600-1,800 mg) may be beneficial for symptoms, but showed no significant change in nerve conduction velocity. This conflicts with the Cochrane review above; the evidence is mixed.

4. Csiha S, Hernyak M, Molnar A, et al. (2025). Alpha-Lipoic Acid Treatment Reduces Advanced Glycation End Products in Type 2 Diabetic Patients with Neuropathy. *Biomedicines* 13(2):438.

PubMed · PMID 40002851 ■

Small single-group biomarker study (54 people with neuropathy on ALA, compared with 24 people without neuropathy): AGE levels were modestly lower after 6 months. Not placebo-controlled.

5. Vallianou N, Evangelopoulos A, Koutalas P. (2009). Alpha-Lipoic Acid and Diabetic Neuropathy. *The Review of Diabetic Studies* 6(4):230-236.

PubMed · PMID 20043035 ■

Review of ALA in diabetic neuropathy. A separate 2019 meta-analysis of 10 RCTs (553 people with uncomplicated type 2 diabetes) found no difference from placebo in HbA1c or fasting glucose (Ebada MA, et al., Iran J Pharm Res).

CHAPTER 4

Omega-3 & Inflammation: What the Research Shows

Fish-oil fats and markers of inflammation — evidence is from mixed populations, not specifically prediabetes.

Blood sugar is only part of the picture. Researchers have also studied low-grade, chronic inflammation — sometimes called **meta-inflammation** — and its links to insulin resistance, involving markers like TNF-alpha, IL-6 and CRP. Omega-3 fats (EPA and DHA, found in fatty fish and fish oil) are among the nutrients studied for their effect on these markers; they are not the only one, and the evidence is mixed in quality.

01. What Omega-3 May Do

Omega-3s are converted in the body into **specialized pro-resolving mediators (SPMs)** — resolvins, protectins and maresins — molecules thought to help switch off inflammation, rather than just blocking pain signals the way common anti-inflammatories do. Whether supplements reliably raise SPM levels in people is still being studied. A 2022 umbrella review of 32 meta-analyses found modest average reductions in CRP, TNF-alpha and IL-6 with omega-3 supplementation, with high variability between studies and across many different health conditions, not specifically prediabetes. [2]

CONTEXT: REDUCE-IT (PRESCRIPTION OMEGA-3)

"The REDUCE-IT trial tested prescription icosapent ethyl (4 g a day) in statin-treated adults with high cardiovascular risk. It is not the same as over-the-counter fish oil, and it does not show that supplements prevent heart disease."

— Bhatt DL, et al. N Engl J Med 2019 | PMID 30415628

02. Product Quality

Fish oil can **oxidize**, and oxidized fats may promote inflammation rather than reduce it. The research behind this is mostly from intravenous fish oil in hospital settings, not oral supplements, but the general quality concern is a reasonable one. [5] Third-party testing (such as IFOS) checks oxidation levels; look for a product that publishes its test results if you decide to use one.

METABOLIC REPAIR DOSE

High FBS & Elevated Triglycerides

A higher dose sometimes used in trials. Discuss with your doctor first, especially if you take blood thinners.

up to 2,000–4,000 mg / day

(doctor-guided)

MAINTENANCE DOSE

Long-Term Protection

A common general maintenance dose; check the EPA+DHA content per serving on the label.

1,000–2,000 mg / day

⚠ PROTOCOL NOTE: THE FATTY MEAL RULE

Omega-3s are fat-soluble, so taking them with a meal that has some fat may help absorption.

PEER-REVIEWED REFERENCES · PUBMED / PMC VERIFIED

1. Khalili L, Valdes-Ramos R, Harbige LS. (2021). Effect of n-3 (Omega-3) Polyunsaturated Fatty Acid Supplementation on Metabolic and Inflammatory Biomarkers and Body Weight in Patients with Type 2 Diabetes Mellitus: A Systematic Review and Meta-Analysis of RCTs. *Metabolites* 11(11):742.

PMC Free Full Text · PMC8620218 ■

30 RCTs in people with type 2 diabetes: omega-3 improved several glycemic and lipid markers on average. This population already has diabetes; results may not carry over to prediabetes.

2. Kavyani Z, Musazadeh V, Fathi S, Faghfouri AH, Dehghan P, Sarmadi B. (2022). Efficacy of the Omega-3 Fatty Acids Supplementation on Inflammatory Biomarkers: An Umbrella Meta-Analysis. *International Immunopharmacology* 111:109104.

PubMed · PMID 35914448 ■

32 meta-analyses: modest average reductions in CRP, TNF-alpha and IL-6, with high heterogeneity between studies; populations studied had various conditions, and the authors describe omega-3 as an adjuvant (add-on), not a primary treatment.

3. Calder PC. (2017). Omega-3 Fatty Acids and Inflammatory Processes: From Molecules to Man. *Biochemical Society Transactions* 45(5):1105-1115.

PubMed · PMID 28900017 ■

Narrative review: EPA/DHA partly inhibit several aspects of the inflammatory response; human trial evidence of benefit is strongest in rheumatoid arthritis and in stabilising atherosclerotic plaques, not specifically prediabetes.

4. Bhatt DL, Steg PG, Miller M, et al.; REDUCE-IT Investigators. (2019). Cardiovascular Risk Reduction with Icosapent Ethyl for Hypertriglyceridemia. *New England Journal of Medicine* 380:11-22.

PubMed · PMID 30415628 ■

Tested prescription icosapent ethyl (4 g/day), a purified EPA drug, in statin-treated adults with elevated triglycerides and high cardiovascular risk. It reduced cardiovascular events in this specific population. It is not the same product as over-the-counter fish oil, and its 25% relative risk reduction should not be read as applying to supplements or to prediabetes.

5. Torrinhas RS, Calder PC, et al. (2024). Parenteral Fish Oil: An Update on Evidence and Clinical Use. *Nutrients*.

see publisher record ■

This source discusses INTRAVENOUS fish-oil lipid emulsions used in hospital nutrition, not oral supplements. It is included here only as general background on fish-oil oxidation and does not support oral-supplement dosing claims.

CHAPTER 5 · NUTRIENT RESEARCH

CHAPTER 5

Vitamin D3 & K2: What the Research Shows

Real trial evidence for vitamin D in prediabetes, and where K2 fits in.

Vitamin D acts on a receptor found in many tissues, including the pancreas, and researchers have studied whether vitamin D status affects insulin secretion and the course of prediabetes. The 2024 Endocrine Society guideline recommends vitamin D supplementation for adults with prediabetes at doses above the standard daily allowance, based on this evidence.

01. The Prediabetes Trial Evidence

A 2025 meta-analysis pooled 10 randomized trials (4,478 people) and found that adults with prediabetes taking vitamin D supplements were modestly more likely to return to normal glucose regulation than those on placebo (18.5% vs. 14.0%; about 27% relative increase). [1] This is a real, recent finding, and the Endocrine Society's recommendation above is partly based on it.

WHAT THE D2D SECONDARY ANALYSIS FOUND (SUMMARY)

"In adults with prediabetes taking 4,000 IU of vitamin D3 a day, insulin-related measures did not improve overall. A benefit appeared only among people whose starting vitamin D level was low (below 12 ng/mL)."

— Summarised from Rasouli N, et al. J Clin Endocrinol Metab 2022 | PMID 34473295

02. Where Vitamin K2 Fits

Vitamin K2 (as MK-7) is involved in directing calcium toward bone rather than soft tissue, and some small trials have combined it with vitamin D3 and looked at glucose along with bone markers. A 40-person double-blind trial found combined D3+K2 lowered fasting glucose more than either alone, but this is one small study and has not been widely replicated. [4]

METABOLIC REPAIR DOSE

Vitamin D3

A dose at or above the Endocrine Society's suggested range for prediabetes. A blood test (25-OH vitamin D) before and during supplementation is recommended; very high doses over long periods can cause harm, so check with your doctor.

1,000–4,000 IU / day (doctor-guided)

K2 SYNERGY DOSE

Vitamin K2 (MK-7 Form)

Sometimes paired with D3; evidence specific to glucose is limited to small studies. Take with a meal containing some fat.

90–180 mcg / day

⚠ PROTOCOL NOTE: THE FAT-SOLUBILITY RULE

Vitamin D3 and K2 are fat-soluble, so taking them with a meal that has some fat may help absorption. Very high vitamin D intakes over long periods can be harmful; stay within your doctor's recommended range.

PEER-REVIEWED REFERENCES · PUBMED / PMC VERIFIED

1. Kretschmer PM, Balk EM, Pittas AG. (2025). Effect of Vitamin D on Regression to Normal Glucose Regulation in Adults With Prediabetes. *Journal of the Endocrine Society* 9(5):bvaf042.

PubMed · PMID 40144812 ■ | PMC Free Full Text · PMC11938431 ■

10 RCTs, 4,478 people with prediabetes: 18.5% on vitamin D vs. 14.0% on placebo returned to normal glucose regulation (RR 1.27, 95% CI 1.12-1.45). Confirmed accurate.

2. Rasouli N, Brodsky IG, Chatterjee R, Kim SH, Pratley RE, Staten MA, Pittas AG; D2d Research Group. (2022). Effects of Vitamin D Supplementation on Insulin Sensitivity and Secretion in Prediabetes. *Journal of Clinical Endocrinology & Metabolism* 107(1):230-240.

PubMed · PMID 34473295 ■

4,000 IU/day for 24 months did NOT improve an index of beta-cell function overall. A benefit was seen only in the subgroup with baseline vitamin D below 12 ng/mL. The overall null result is the headline finding; this is a secondary, prespecified subgroup analysis.

3. Jadhav N, Ajgaonkar S, Mehta D, et al. (2022). Molecular Pathways and Roles for Vitamin K2-7 as a Health-Beneficial Nutraceutical. *Frontiers in Pharmacology.*

see publisher record — NOT independently re-verified ■

A mechanistic review of proposed K2 pathways. Not independently re-verified for this edition; the specific insulin-sensitivity mechanism described is not established in human trials.

4. Aguayo-Ruiz JI, et al. (2020). Effect of Supplementation with Vitamins D3 and K2 on Undercarboxylated Osteocalcin and Insulin Serum Levels in Patients with Type 2 Diabetes Mellitus. *Diabetology & Metabolic Syndrome.*

see publisher record — NOT independently re-verified ■

Small trial (40 people with type 2 diabetes, not prediabetes). Not independently re-verified for this edition; treat as preliminary.

5. Xie C, Gong J, Zheng C, et al. (2024). Effects of Vitamin K Supplementation on Bone Mineral Density and Bone Metabolism. *Bone & Joint Research.*

see publisher record ■

This source is about BONE density, not glucose or arterial calcium. It is kept here only as a general K2 reference and does not support any claim about blood sugar.

CHAPTER 6 · EMERGING RESEARCH

CHAPTER 6

Akkermansia muciniphila: An Early-Stage Gut Supplement

A single small trial found a benefit only in a subgroup — promising, but early.

Gut bacteria influence metabolism in ways researchers are still mapping out. One species — *Akkermansia muciniphila* — has drawn interest for its possible link to insulin sensitivity and weight. Human trial evidence is still limited, and the most recent and largest trial found a benefit only in a specific subgroup, not overall.

01. A Proposed Mechanism

One proposed mechanism: low *Akkermansia* levels may be linked to a thinner gut lining, which in animal studies allowed more bacterial byproducts (LPS) into the bloodstream and was associated with inflammation and insulin resistance. The 2013 study behind this finding was in mice, not people. [5] Whether the same pathway matters in humans is not established. [4, 5]

KEY FINDING (CELL METABOLISM 2025)

"In this 12-week trial in adults with type 2 diabetes, both groups lost weight and HbA1c with lifestyle guidance, and the supplement did not significantly outperform placebo overall. Only people with low starting Akkermansia showed extra benefit."

— Summarised from Zhang Y, et al. Cell Metab 2025 | PMID 39879980

02. Pasteurized vs. Live Forms

The first human trial (Depommier et al., *Nature Medicine*, 2019) was a small, 3-month pilot in 32 overweight or obese adults with insulin resistance, and reported that pasteurized (heat-treated) *A. muciniphila* improved several metabolic measures more than placebo. [1] It did not test live cells head-to-head within the same trial, so 'outperformed live cells' is not something this single study shows. A 2025 laboratory study found that an *A. muciniphila* extract increased GLP-1 release from human cells in a dish; this is a cell-culture finding, not a result in people. [3]

METABOLIC PROTOCOL

Pasteurized A. muciniphila

This is an early-stage, lightly studied supplement. If you and your doctor decide to try it, look for a refrigerated, reputable product; evidence of an added benefit beyond diet and movement is limited to one subgroup in one trial.

per label (follow product instructions)

POLYPHENOL SYNERGY

Prebiotic Feeding

Polyphenol-rich foods (berries, pomegranate, green tea) are a reasonable part of a healthy diet regardless of any effect on gut bacteria; evidence they boost Akkermansia specifically in people is limited.

food-based (no established supplement dose)

⚠ PROTOCOL NOTE: EMPTY STOMACH TIMING

Unlike most probiotics, *Akkermansia* is best taken on an empty stomach (30 min before breakfast) to ensure transit into the lower intestinal mucus layer where it exerts barrier-strengthening and GLP-1 effects. Avoid taking with very hot beverages which can damage the pasteurized bacteria.

PEER-REVIEWED REFERENCES · PUBMED / PMC VERIFIED

1. Depommier C, Everard A, Druart C, et al. (2019). Supplementation with Akkermansia muciniphila in Overweight and Obese Human Volunteers: A Proof-of-Concept Exploratory Study. *Nature Medicine* 25:1096-1103.

PubMed · PMID 31263284 ■ | PMC Free Full Text · PMC6699990 ■

Small pilot trial (32 people with overweight/obesity and insulin resistance, 3 months): pasteurized A. muciniphila improved several metabolic measures versus placebo. It is a proof-of-concept study, not a confirmation of benefit.

2. Zhang Y, Liu R, Chen Y, et al. (2025). Akkermansia Supplementation in Overweight/Obese Type 2 Diabetes. *Cell Metabolism* 37(3):592-605.e6.

PubMed · PMID 39879980 ■

12-week trial, 58 people with overweight/obesity and type 2 diabetes: both groups lost weight and HbA1c with lifestyle guidance; NO significant difference between supplement and placebo overall. Only the subgroup with low starting Akkermansia showed an extra benefit.

3. Arukha AP, Nayak S, Swain DM. (2025). Effect of Akkermansia muciniphila on GLP-1 and Insulin Secretion. *Nutrients* 17(15):2516.

see publisher record — NOT independently re-verified ■

A cell-culture (in vitro) study. Findings in cultured cells do not establish an effect in living people and should not be read as evidence of a GLP-1 benefit from supplementation.

4. Ye WY, Cai Y. (2025). Akkermansia muciniphila: Gut Microbiota Crosstalk Against Oxidative Stress and Clinical Prospects. *BMC Microbiology*.

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A review article. Not independently re-verified for this edition.

5. Everard A, Belzer C, Geurts L, et al. (2013). Cross-Talk Between Akkermansia muciniphila and Intestinal Epithelium Controls Diet-Induced Obesity. *Proceedings of the National Academy of Sciences* 110(22):9066-9071.

PubMed · PMID 23671105 ■

A MOUSE study, not a human trial. It supports the proposed mechanism in animals, and does not by itself show the same effect occurs in people.

CONCLUSION

One Way to Sequence These Topics Over 12 Weeks

An example, not a prescription — discuss any plan with your doctor first

This chapter has covered six supplement topics and what the research shows for each. If you and your doctor decide some are worth trying, the table below shows one way to introduce them gradually, starting with the better-evidenced ones. It is an example, not a required sequence, and none of it replaces food, movement, sleep or medical care.

Phase	Weeks	Possible Additions	What the Evidence Covers
Starting Point	1–4	If trying one supplement: berberine (up to 1,500 mg/day, in divided doses) Magnesium (up to 350 mg elemental/day) Chromium (200-600 mcg/day, optional) Omega-3 (1,000-2,000 mg/day)	Evidence discussed in Ch. 1, 2 & 4 — mostly in people with type 2 diabetes, not prediabetes
If Continuing	5–8	Alpha-lipoic acid (300-600 mg/day), if advised Vitamin D3 (1,000-4,000 IU/day) + K2, with a blood test Re-check with your doctor before increasing any dose	Ch. 3 evidence is mainly in diagnosed neuropathy, not prevention. Ch. 5 vitamin D evidence is the strongest in this guide for prediabetes specifically.
Optional, Early-Stage	9–12	Akkermansia (pasteurized), only if you and your doctor decide to Follow the product label for dosing Reassess fasting glucose & HbA1c with your doctor	Ch. 6 evidence is from one small trial with a subgroup benefit — treat as exploratory

Summary of Topics Covered

Ch. 1	**Berberine**	up to 1,500 mg/day	Studied mostly in type 2 diabetes
Ch. 2	**Magnesium + Chromium**	350 mg Mg + 200-600 mcg Cr	Modest effects; mixed evidence
Ch. 3	**Alpha-Lipoic Acid**	300–600 mg/day	Mainly studied for neuropathy
Ch. 4	**Omega-3 EPA/DHA**	1,000–2,000 mg/day	Modest effect on inflammation markers
Ch. 5	**Vitamin D3 + K2**	1,000-4,000 IU + 90-180 mcg	Strongest prediabetes-specific evidence here
Ch. 6	**Akkermansia (pasteurized)**	per label	Early-stage; one subgroup finding

This table summarizes topics discussed in this guide, drawn from the research cited in each chapter. It is not a prescription. Reassess fasting glucose, HbA1c and lipids with your doctor periodically, and adjust or stop any supplement in consultation with them.

Master Bibliography

These 30 citations correspond to the in-text references within each chapter, with the corrections and caveats noted in each chapter's own reference list above. Several sources were not independently re-verified for this edition (marked in-chapter); treat those as unconfirmed rather than established.

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Individual results vary and are not guaranteed. Always consult your doctor or a qualified healthcare provider before making changes to your diet, exercise routine, medications, or supplement regimen — especially before starting berberine, alpha-lipoic acid, vitamin D3, omega-3, or any other supplement discussed here. **Do not use these supplements if you are pregnant, planning a pregnancy, or breastfeeding, and do not give them to children.** They can interact with other medicines, including blood thinners, blood pressure medicines and diabetes medicines, and may not suit people with kidney or liver disease.

Evidence quality varies widely between the topics in this guide; several chapters note mixed, limited, or early-stage research, and some cited sources could not be independently re-verified for this edition. Never disregard professional medical advice or delay seeking it because of something you read in this guide.

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